

APPENDIX

A. Proof of Lemma 1

Proof. The matrix $\mathbf{M}_{i_k}$ can be rewritten in a symmetric form

$$\begin{aligned}\mathbf{M}_{i_k} &= \left(\underline{\mathbf{F}}_{i_k} + \mathbf{H}_{i_k,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{i_k,k}^H \right)^{-1} \mathbf{H}_{i_k,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{i_k,k}^H \underline{\mathbf{F}}_{i_k}^{-1} \\ &= \left(\underline{\mathbf{F}}_{i_k} + \mathbf{H}_{i_k,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{i_k,k}^H \right)^{-1} \left(\underline{\mathbf{F}}_{i_k} + \mathbf{H}_{i_k,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{i_k,k}^H - \underline{\mathbf{F}}_{i_k} \right) \underline{\mathbf{F}}_{i_k}^{-1} \\ &= \underline{\mathbf{F}}_{i_k}^{-1} - \left(\underline{\mathbf{F}}_{i_k} + \mathbf{H}_{i_k,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{i_k,k}^H \right)^{-1}.\end{aligned}\tag{11}$$

Let $\underline{\mathbf{N}}_{j_l} = \sum_{(i,k) \neq (j,l)} \mathbf{H}_{j_l,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{j_l,k}^H \succeq \mathbf{0}$, according to matrix inversion lemma, we have

$$\begin{aligned}\mathbf{A}_k &= \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \mathbf{M}_{j_l} \mathbf{H}_{j_l,k} \\ &= \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \left(\underline{\mathbf{F}}_{j_l}^{-1} - \left(\underline{\mathbf{F}}_{j_l} + \mathbf{H}_{j_l,l} \underline{\mathbf{W}}_{j_l} \underline{\mathbf{W}}_{j_l}^H \mathbf{H}_{j_l,l}^H \right)^{-1} \right) \mathbf{H}_{j_l,k} \\ &\preceq \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \underline{\mathbf{F}}_{j_l}^{-1} \mathbf{H}_{j_l,k}, \\ &= \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H (\sigma_{j_l}^2 \mathbf{I} + \underline{\mathbf{N}}_{j_l})^{-1} \mathbf{H}_{j_l,k} \\ &= \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \left(\frac{1}{\sigma_{j_l}^2} \mathbf{I} - \frac{1}{\sigma_{j_l}^2} \underline{\mathbf{N}}_{j_l}^{1/2} \left(\mathbf{I} + \frac{1}{\sigma_{j_l}^2} \underline{\mathbf{N}}_{j_l} \right)^{-1} \underline{\mathbf{N}}_{j_l}^{1/2} \frac{1}{\sigma_{j_l}^2} \right) \mathbf{H}_{j_l,k} \\ &\preceq \sum_{(j,l)} \frac{\alpha_{j_l}}{\sigma_{j_l}^2} \mathbf{H}_{j_l,k}^H \mathbf{H}_{j_l,k} \preceq \sum_{(j,l)} \frac{\alpha_{j_l}}{\sigma_{j_l}^2} \lambda_{\max}(\mathbf{H}_{j_l,k}^H \mathbf{H}_{j_l,k}) \mathbf{I} = \left(\sum_{(j,l)} \frac{\alpha_{j_l}}{\sigma_{j_l}^2} \|\mathbf{H}_{j_l,k}\|_2^2 \right) \mathbf{I},\end{aligned}$$

for $k = 1, \dots, k$. Denote $L = \max_k \sum_{(j,l)} \frac{2\alpha_{j_l}}{\sigma_{j_l}^2} \|\mathbf{H}_{j_l,k}\|_2^2$, and the Hessian of the WSR can be bounded as

$$\nabla^2 f(\{\underline{\mathbf{W}}_{i_k}\}) \succeq -L\mathbf{I},\tag{12}$$

for all $\underline{\mathbf{W}}_{i_k} \in \mathbb{C}^{N_t \times N_s}$, which implies the gradient of f is L -Lipchitz. $\square$

B. Proof of Lemma 3

Proof. Let $\tilde{\mathbf{F}}_{i_k} = \mathbf{F}_{i_k} + \mathbf{H}_{i_k,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{i_k,k}^H = \sum_{(j,l)} \mathbf{H}_{i_k,l} \underline{\mathbf{W}}_{j_l} \underline{\mathbf{W}}_{j_l}^H \mathbf{H}_{i_k,l}^H + \sigma_{i_k}^2 \mathbf{I}$, the gradient $\nabla_{\underline{\mathbf{W}}_{i_k}} f(\{\underline{\mathbf{W}}_{i_k}\}) = \mathbf{G}_{i_k}$ can be rewritten as follows:

$$\begin{aligned}\mathbf{G}_{i_k} &= \left(\alpha_{i_k} \mathbf{H}_{i_k,k}^H \underline{\mathbf{F}}_{i_k}^{-1} \mathbf{H}_{i_k,k} - \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \left(\underline{\mathbf{F}}_{j_l}^{-1} - \tilde{\mathbf{F}}_{j_l}^{-1} \right) \mathbf{H}_{j_l,k} \right) \underline{\mathbf{W}}_{i_k} \\ &= \left(\sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \tilde{\mathbf{F}}_{j_l}^{-1} \mathbf{H}_{j_l,k} - \sum_{(j,l) \neq (i,k)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \underline{\mathbf{F}}_{j_l}^{-1} \mathbf{H}_{j_l,k} \right) \underline{\mathbf{W}}_{i_k}.\end{aligned}\tag{13}$$

Considering the sum of the inner products between $\mathbf{G}_{i_k}$ and $\underline{\mathbf{W}}_{i_k}$, i.e.

$$\begin{aligned}\sum_{(i,k)} \text{tr}(\underline{\mathbf{W}}_{i_k}^H \mathbf{G}_{i_k}) &= \sum_{(i,k)} \text{tr} \left(\underline{\mathbf{W}}_{i_k}^H \left(\sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \tilde{\mathbf{F}}_{j_l}^{-1} \mathbf{H}_{j_l,k} - \sum_{(j,l) \neq (i,k)} \alpha_{j_l} \mathbf{H}_{j_l,k}^H \underline{\mathbf{F}}_{j_l}^{-1} \mathbf{H}_{j_l,k} \right) \underline{\mathbf{W}}_{i_k} \right) \\ &= \sum_{(j,l)} \alpha_{j_l} \text{tr} \left(\sum_{(i,k)} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{j_l,k}^H \tilde{\mathbf{F}}_{j_l}^{-1} \mathbf{H}_{j_l,k} \underline{\mathbf{W}}_{i_k} - \sum_{(i,k) \neq (j,l)} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{j_l,k}^H \underline{\mathbf{F}}_{j_l}^{-1} \mathbf{H}_{j_l,k} \underline{\mathbf{W}}_{i_k} \right) \\ &= \sum_{(j,l)} \alpha_{j_l} \text{tr} \left(\tilde{\mathbf{F}}_{j_l}^{-1} \left(\sum_{(i,k)} \mathbf{H}_{j_l,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{j_l,k}^H \right) - \underline{\mathbf{F}}_{j_l}^{-1} \left(\sum_{(i,k) \neq (j,l)} \mathbf{H}_{j_l,k} \underline{\mathbf{W}}_{i_k} \underline{\mathbf{W}}_{i_k}^H \mathbf{H}_{j_l,k}^H \right) \right).\end{aligned}\tag{14}$$

Since matrices

$$\begin{aligned}\tilde{\mathbf{F}}_{i_k} &= \sum_{(j,l)} \mathbf{H}_{i_k,l} \mathbf{W}_{j_l} \mathbf{W}_{j_l}^H \mathbf{H}_{i_k,l}^H + \sigma_{i_k}^2 \mathbf{I}, \\ \mathbf{F}_{i_k} &= \sum_{(j,l) \neq (i,k)} \mathbf{H}_{i_k,l} \mathbf{W}_{j_l} \mathbf{W}_{j_l}^H \mathbf{H}_{i_k,l}^H + \sigma_{i_k}^2 \mathbf{I},\end{aligned}\tag{15}$$

for $k = 1, \dots, K$, $i = 1, \dots, I_k$, according to matrix inversion lemma, we have

$$\begin{aligned}\sum_{(i,k)} \text{tr}(\mathbf{W}_{i_k}^H \mathbf{G}_{i_k}) &= \sum_{(j,l)} \alpha_{j_l} \sigma_{j_l}^2 \text{tr}(\mathbf{F}_{j_l}^{-1} - \tilde{\mathbf{F}}_{j_l}^{-1}) \\ &= \sum_{(j,l)} \alpha_{j_l} \sigma_{j_l}^2 \text{tr}(\mathbf{F}_{j_l}^{-1} - (\mathbf{F}_{j_l} + \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} \mathbf{W}_{j_l}^H \mathbf{H}_{j_l,l}^H)^{-1}) \\ &= \sum_{(j,l)} \alpha_{j_l} \sigma_{j_l}^2 \text{tr}(\mathbf{F}_{j_l}^{-1} - (\mathbf{F}_{j_l}^{-1} - \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} (\mathbf{I} + \mathbf{W}_{j_l}^H \mathbf{H}_{j_l,l}^H \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l})^{-1} \mathbf{W}_{j_l}^H \mathbf{H}_{j_l,l}^H \mathbf{F}_{j_l}^{-1})) \\ &= \sum_{(j,l)} \alpha_{j_l} \sigma_{j_l}^2 \text{tr}(\mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} (\mathbf{I} + \mathbf{W}_{j_l}^H \mathbf{H}_{j_l,l}^H \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l})^{-1} \mathbf{W}_{j_l}^H \mathbf{H}_{j_l,l}^H \mathbf{F}_{j_l}^{-1}) \\ &\geq 0,\end{aligned}\tag{16}$$

the inequality holds if and only if $\mathbf{H}_{i_k,k} \mathbf{W}_{i_k} = \mathbf{0}$ for all $i = 1, \dots, I_k$, $k = 1, \dots, K$. □